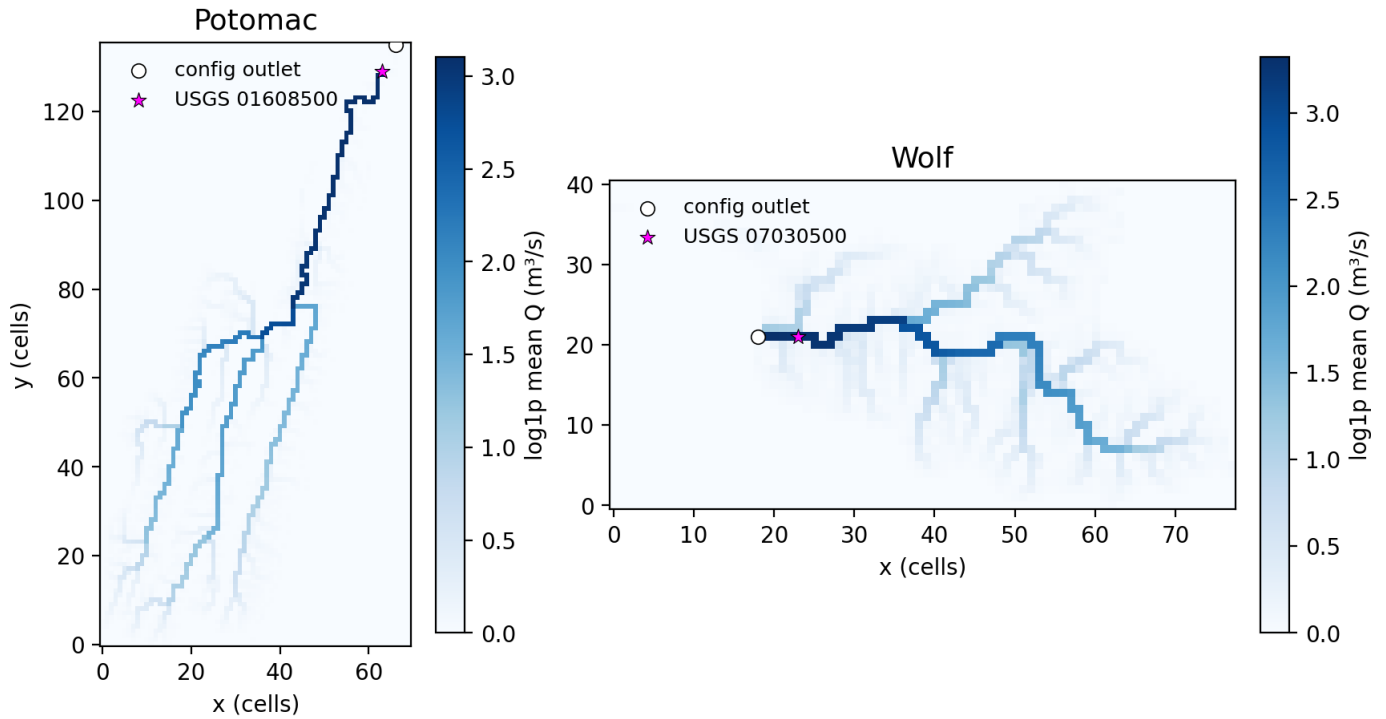


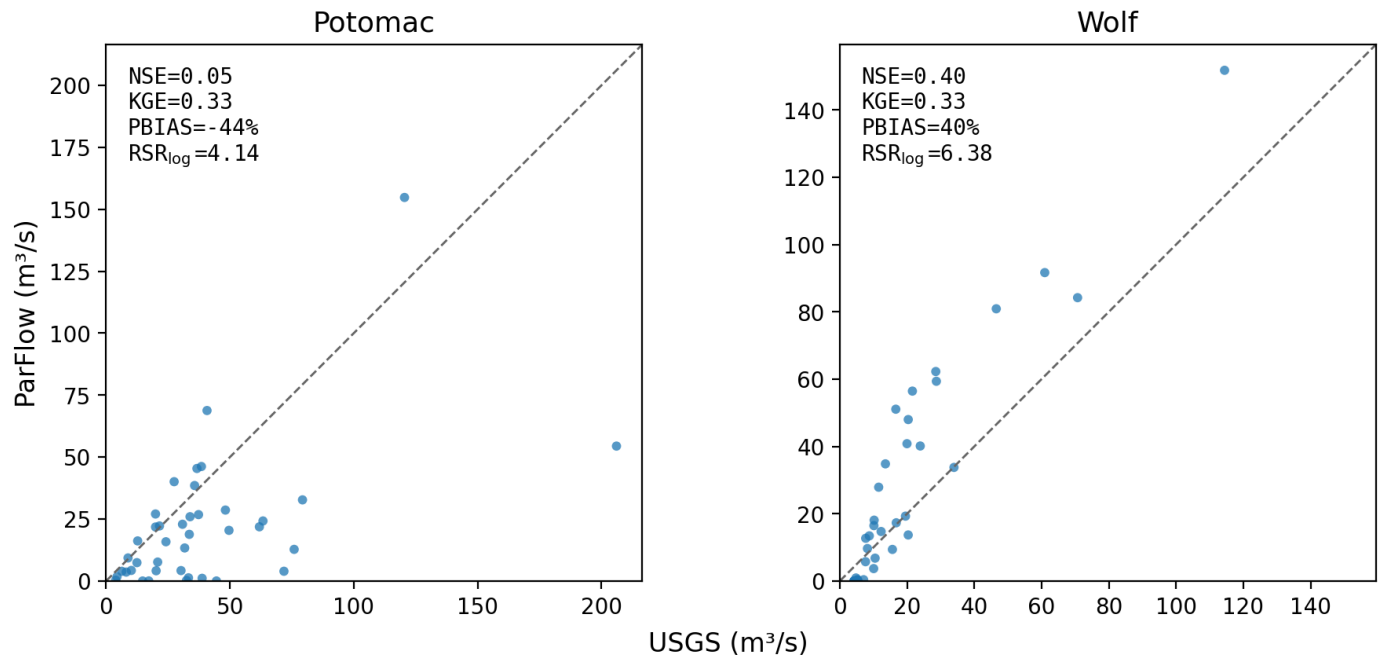
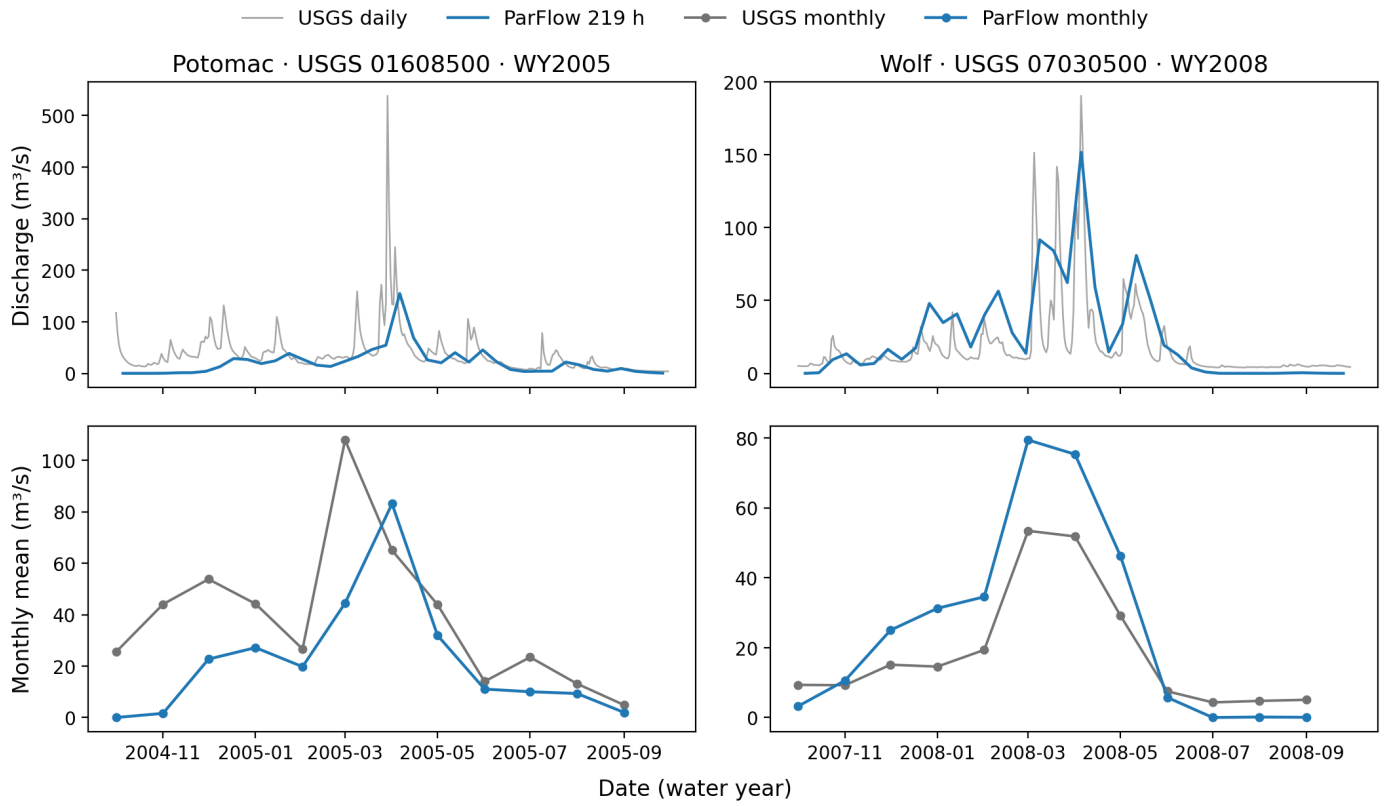
usgs streamflow validation summary

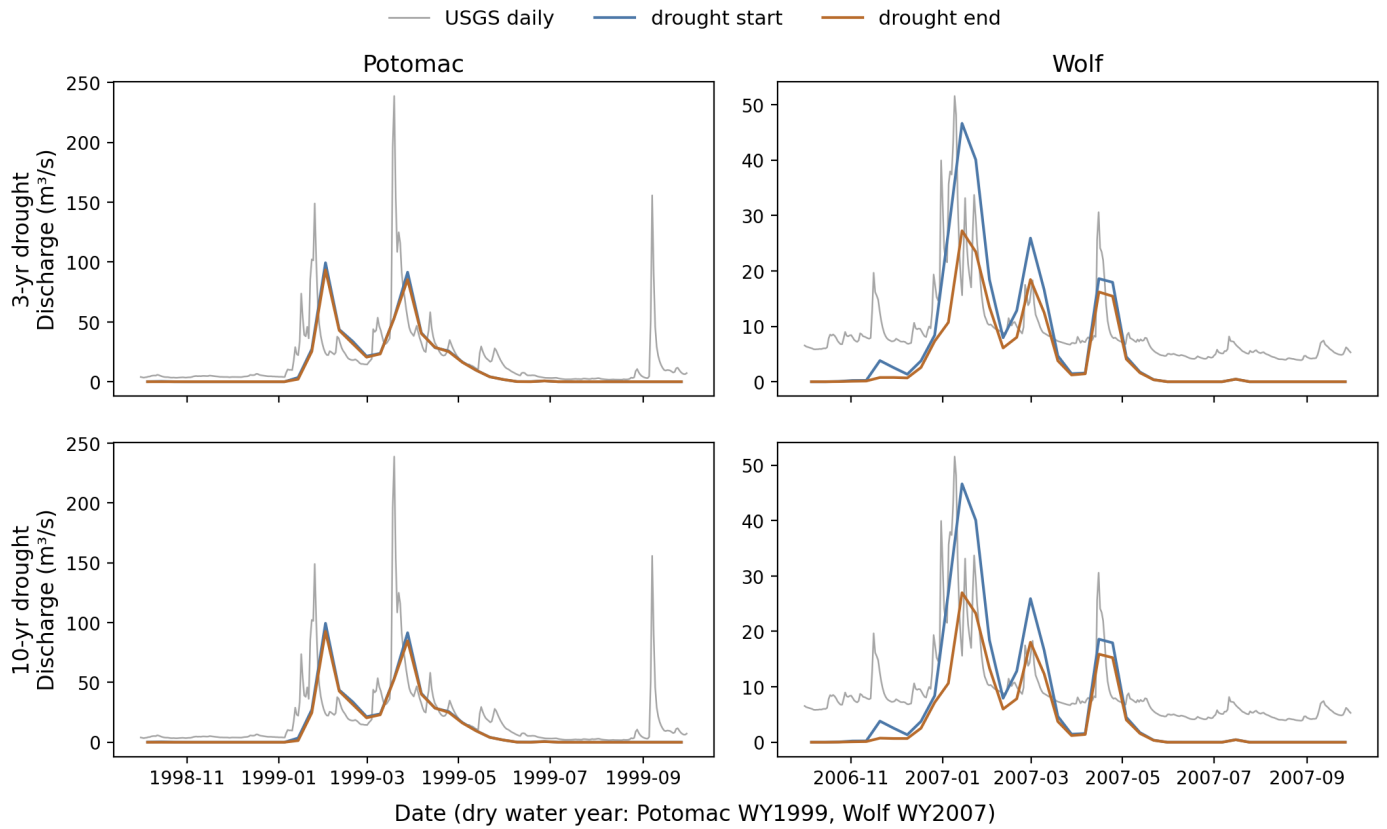
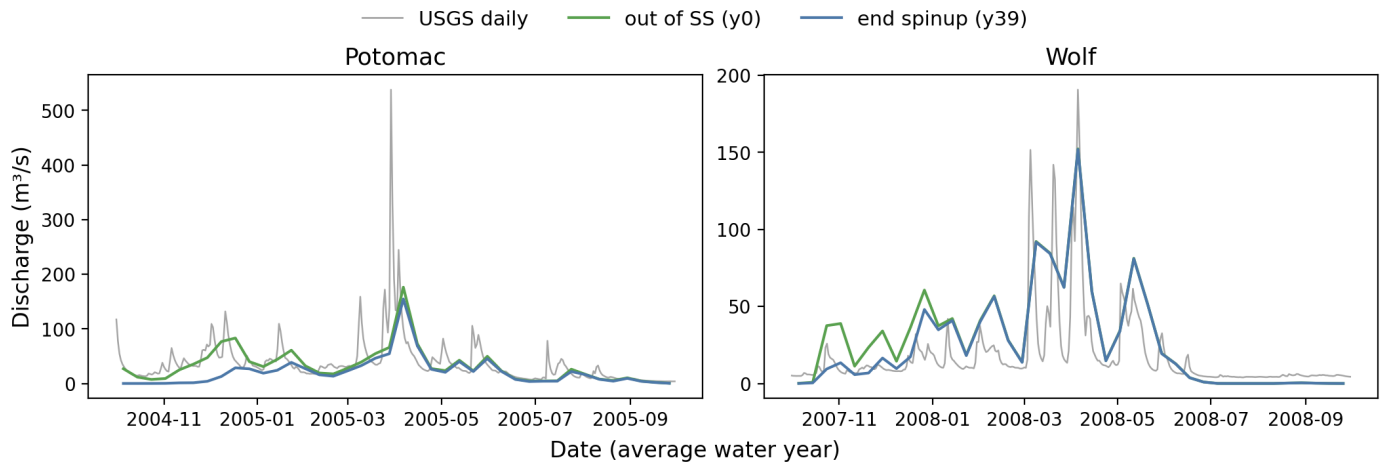
USGS streamflow validation summary

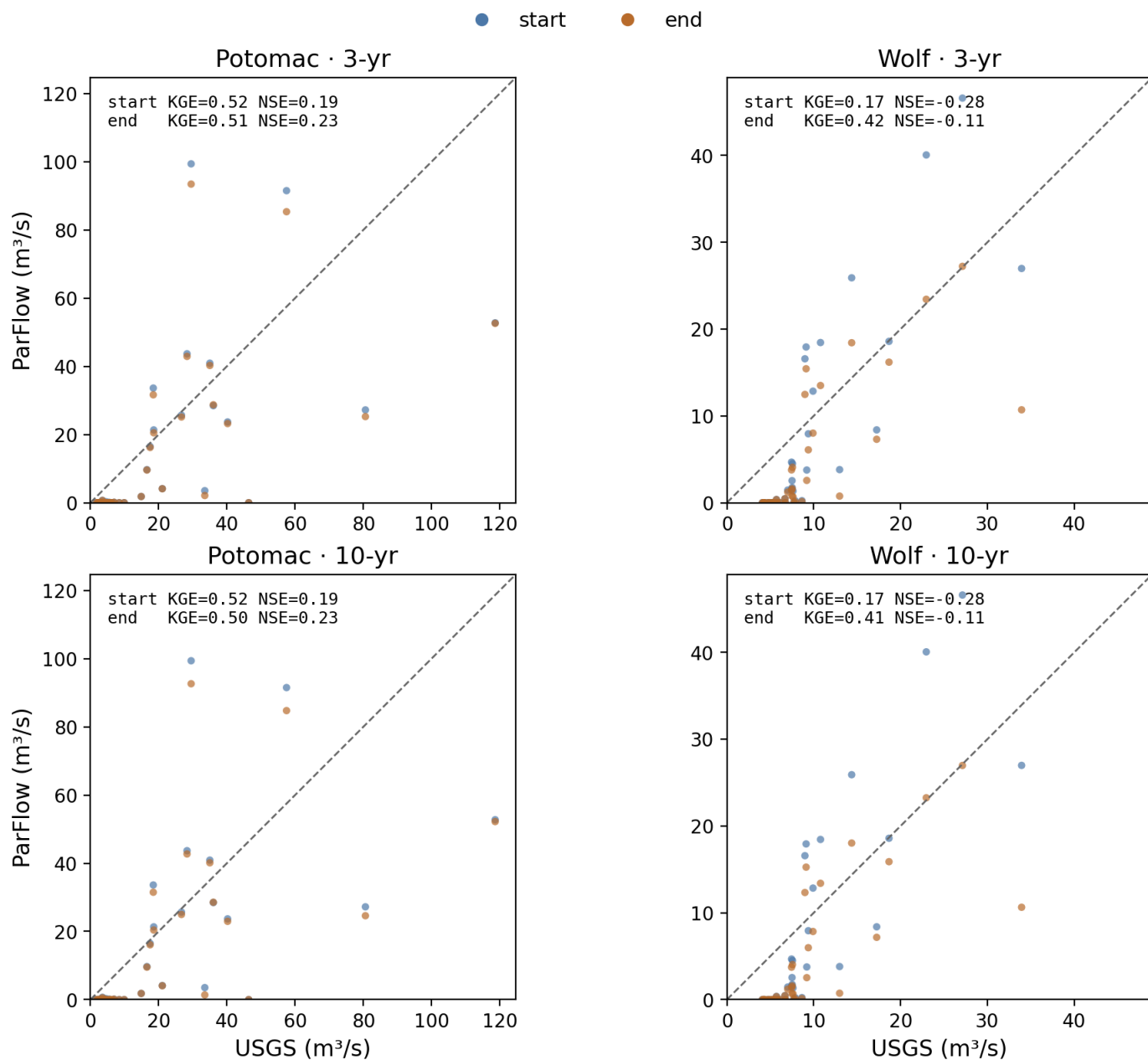
Date: August 2026

Notebook: average_year_usgs_validation.ipynb

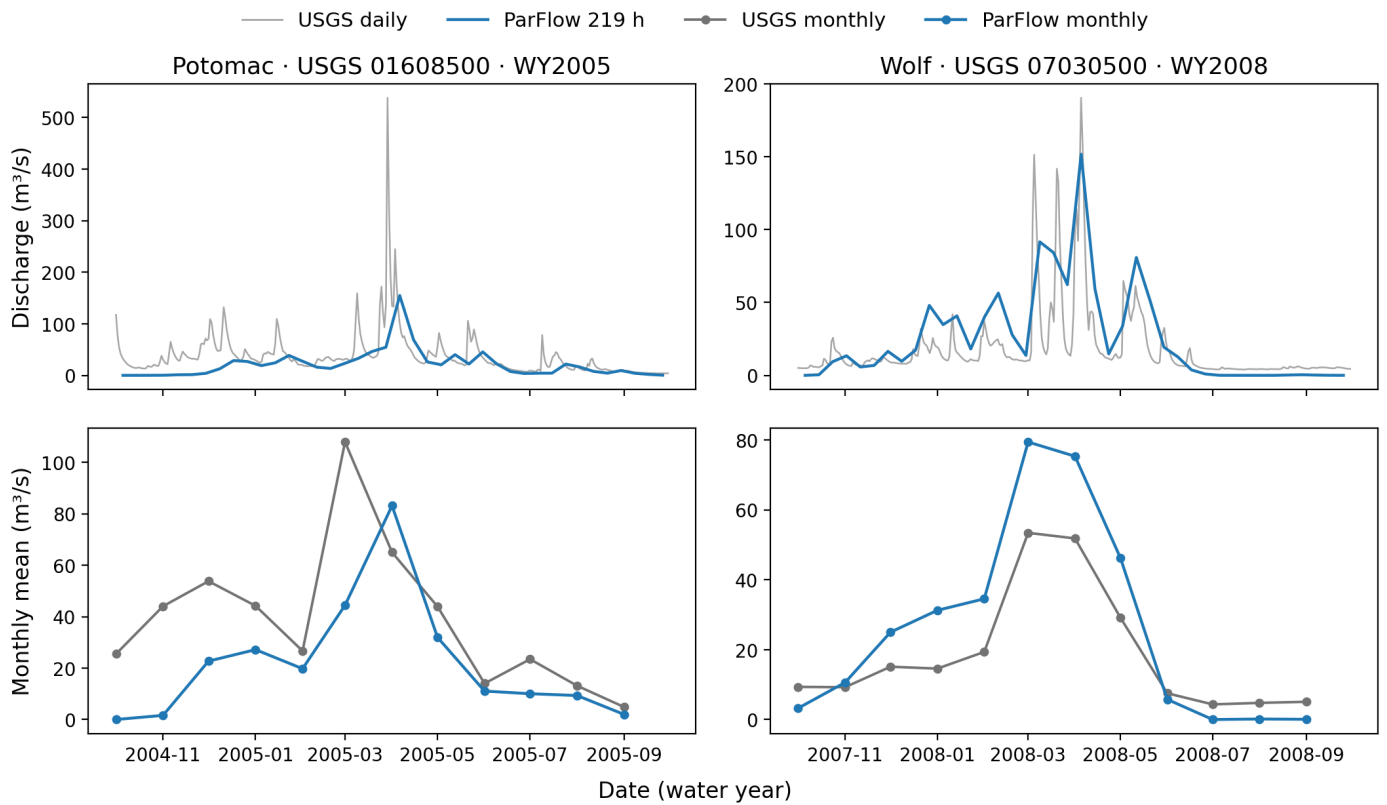
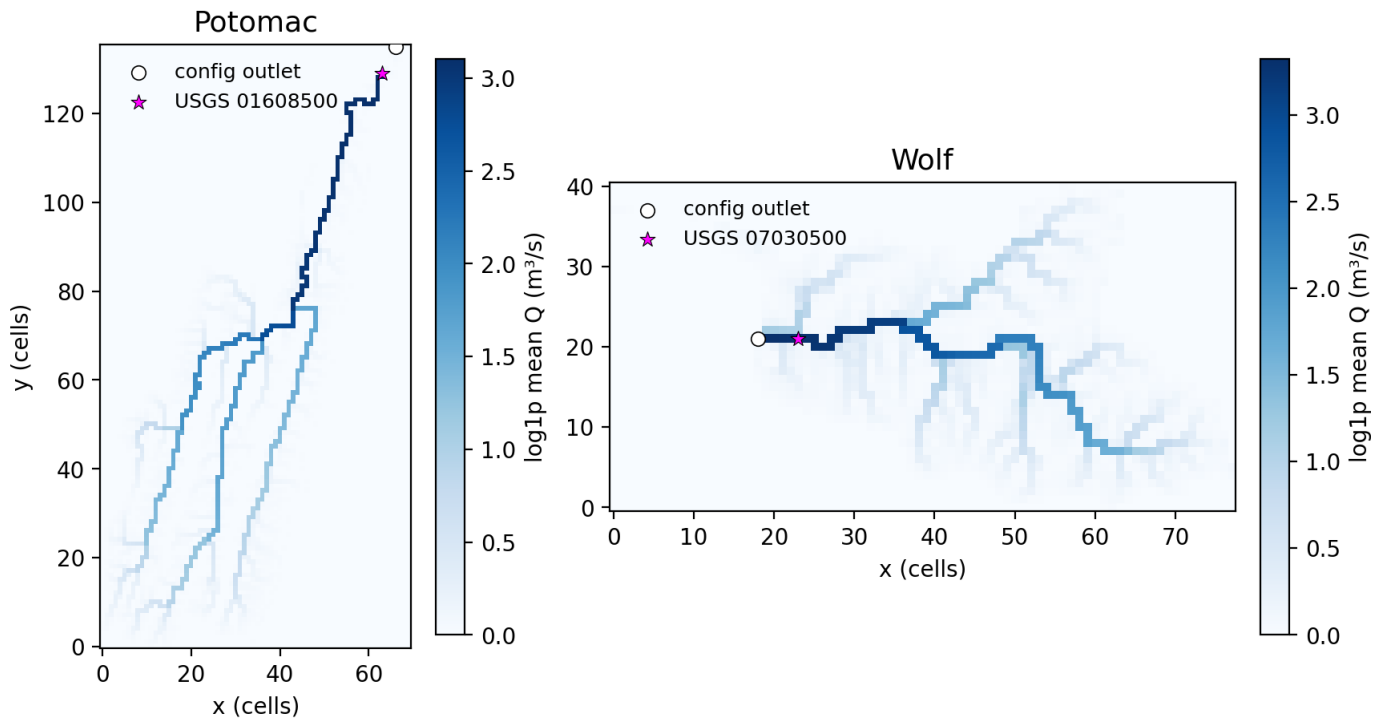


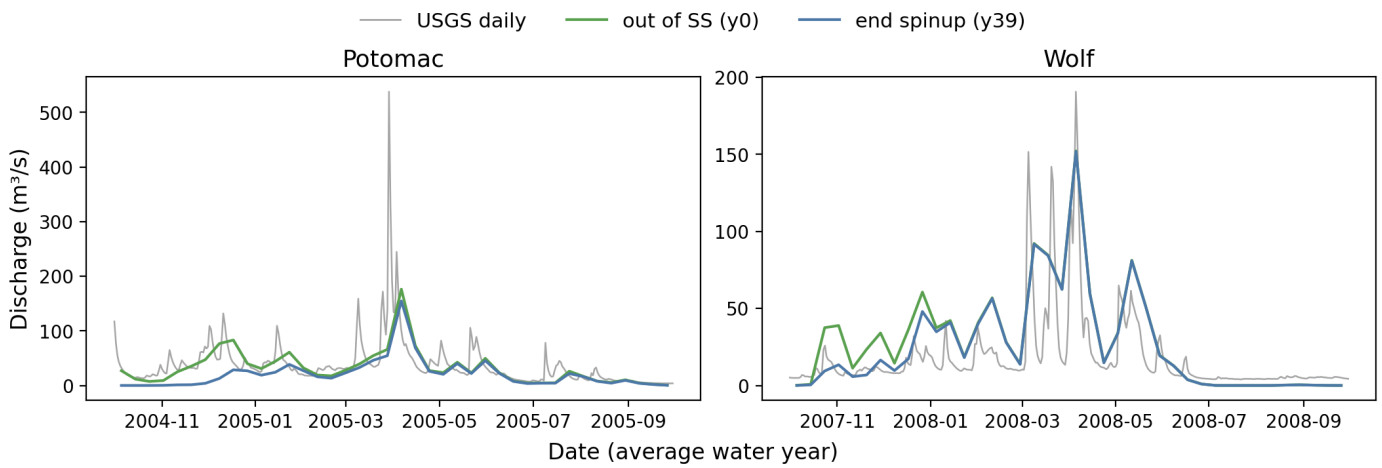
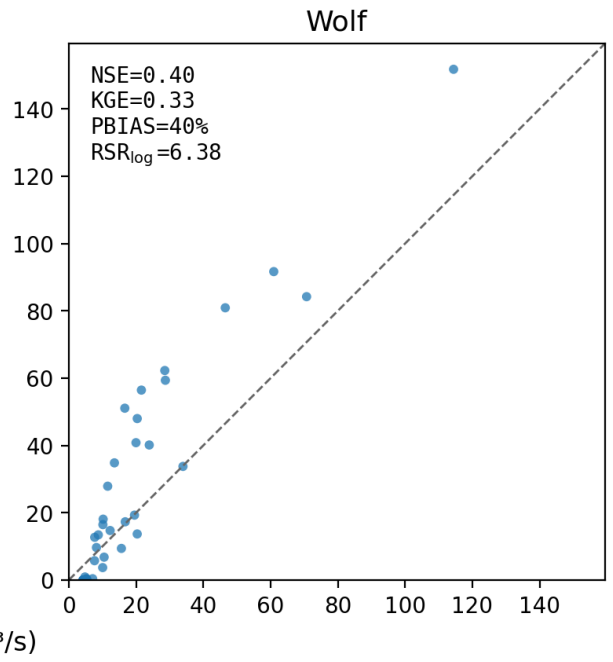
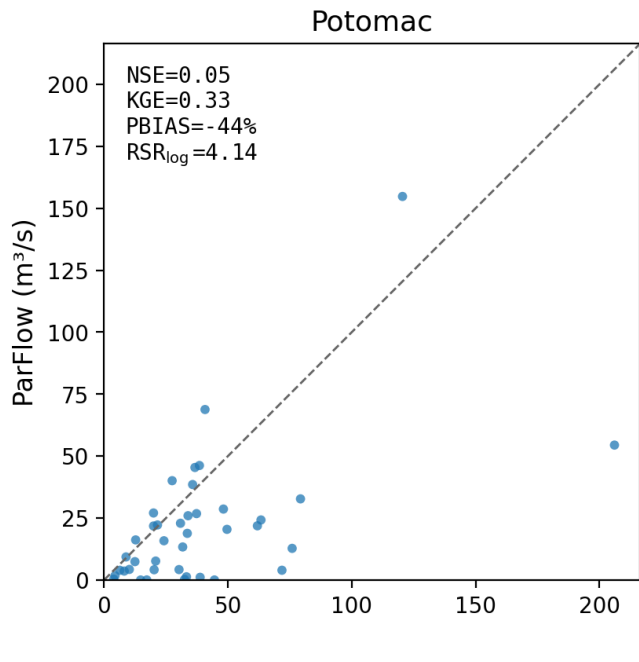


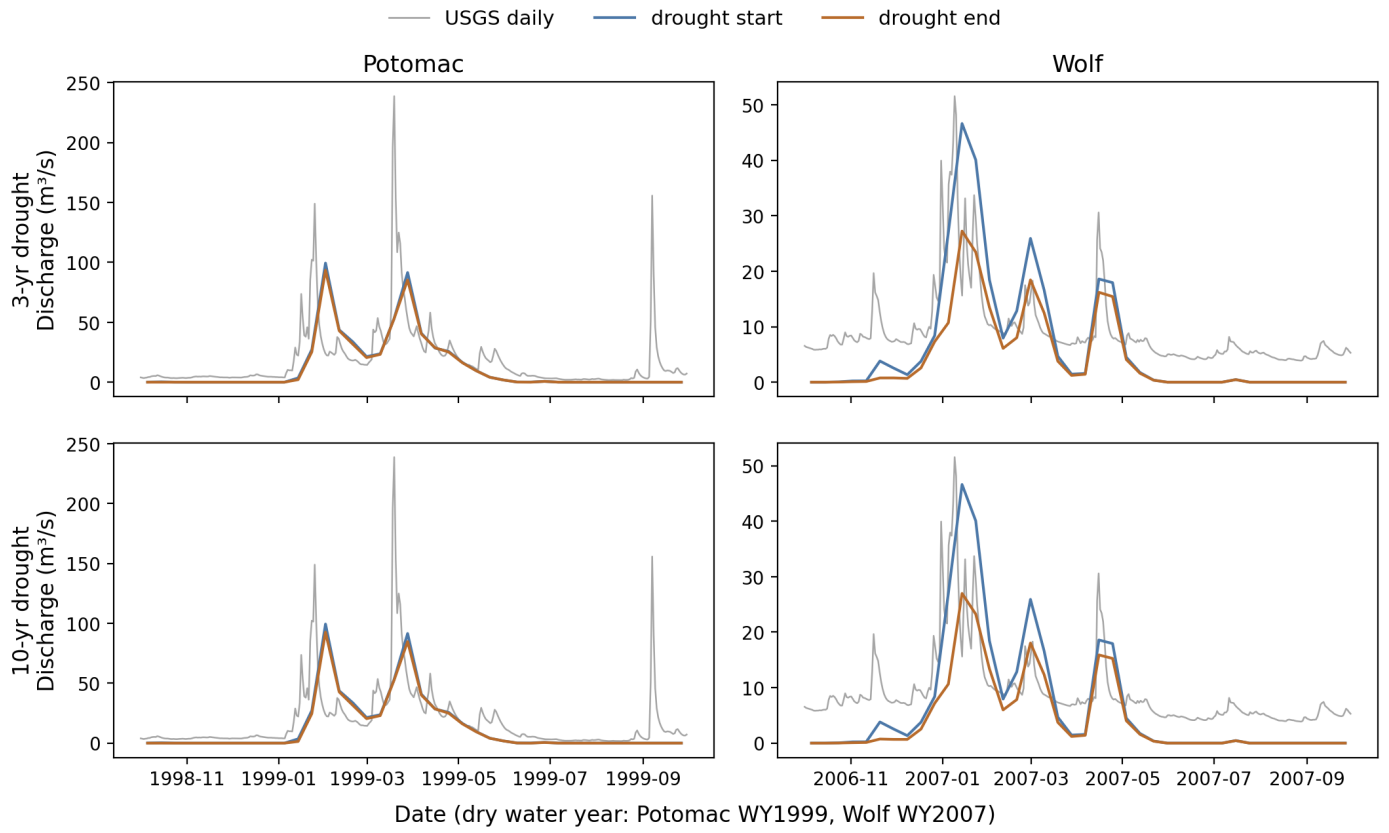


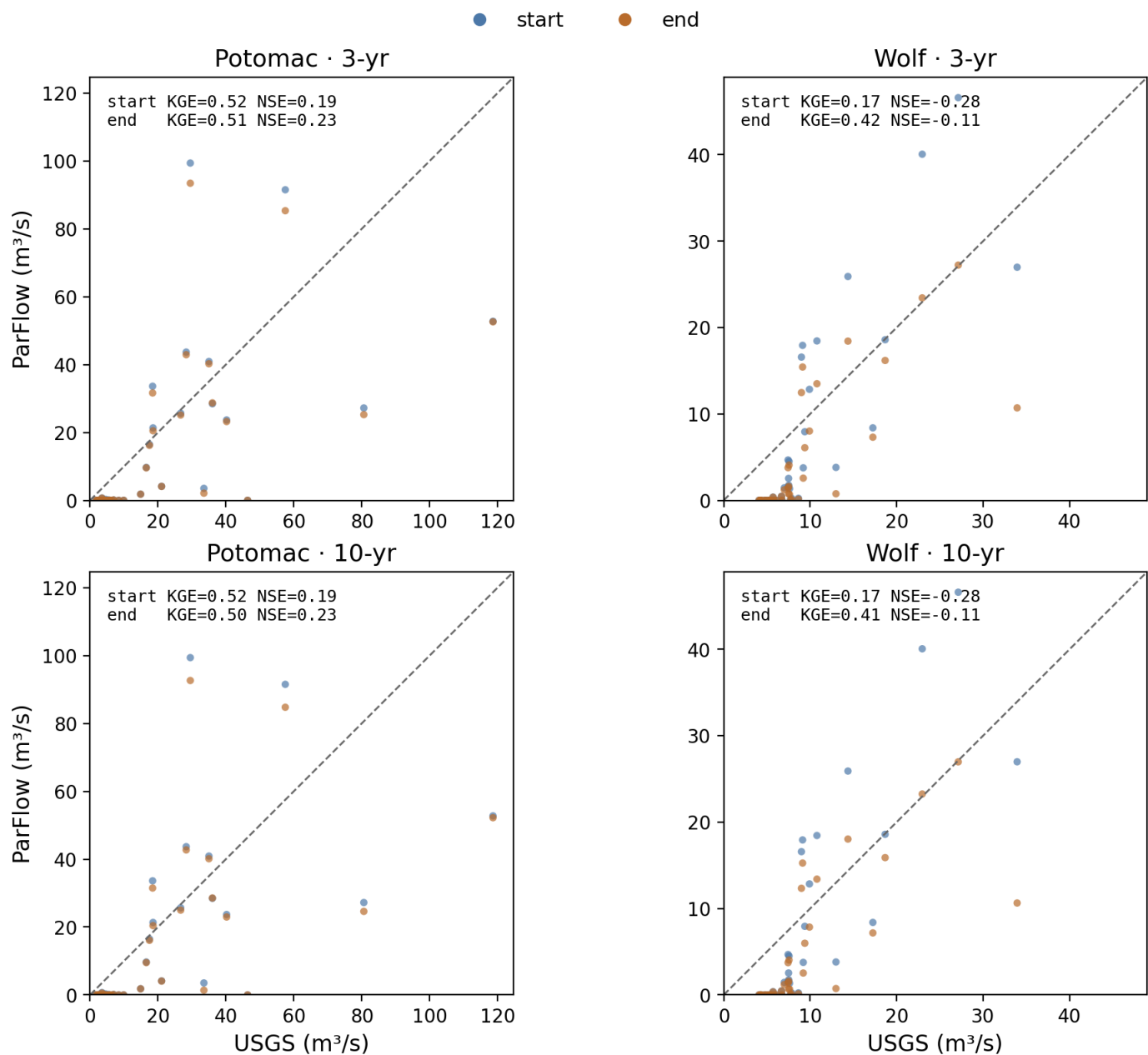


Figures: figures/ (average_year_usgs_*.png, drought_year_usgs_start_vs_end_*.png)









ParFlow overland_flow at drainage-matched USGS gages vs NWIS daily discharge. Model values are 219 h means; USGS is block-averaged onto the same windows. Metrics: NSE, KGE, PBIAS, correlation (r).

Setup

Domain	HUC	Average WY	Dry WY	Validation gage	Why this gage
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Potomac (potomac2)	02070001	2005	1999	01608500 Springfield, WV (~1461 mi2)	Matches basin / outlet scale. Config gage 01607500 Brandywine is
Wolf (wolf2)	08010210	2008	2007	07030500 Rossville, TN (~503 mi2)	only 103 mi2 Config reference gage.

Gage locations are mapped to the 1 km grid and snapped to the nearest channel cell (high mean overland flow). Config outlets are not always on the mainstem (notably Potomac (66, 135)).

Forcing years repeat (same average or dry CW3E water year). Differences between "start" and "end" therefore reflect antecedent storage, not different meteorology.

1. Average year after transient spinup

Post-spinup average year (short_baseline year index 40 ~ 39) vs USGS for the average water year.

Domain	WY	NSE	KGE	PBIAS	r	Sim / obs mean (m3/s)
Potomac	2005	0.05	0.33	-44%	0.56	21.3 / 38.2
Wolf	2008	0.40	0.33	+40%	0.94	25.7 / 18.3

Wolf timing tracks well (high r) but runs high. Potomac is biased low with weaker skill.

2. Average year: out of steady state vs end of spinup

Same average WY and gage; year 0 (first year after steady-state IC) vs year 39 (end of 40-year average spinup).

Domain	Phase	NSE	KGE	PBIAS	Sim mean
Potomac	Out of SS (y0)	0.34	0.60	-16%	32.2 (m3/s)
Potomac	End spinup (y39)	0.05	0.33	-44%	21.3
Wolf	Out of SS (y0)	0.20	0.20	+59%	29.1
Wolf	End spinup (y39)	0.40	0.33	+40%	25.7

d (end - SS): Potomac dNSE -0.29, dKGE -0.27 - skill worse after spinup (mean flow drops). Wolf dNSE +0.20, dKGE +0.13 - skill better (wet SS bias relaxes). Year 39 ~ year 40 for both.

3. Drought (dry) year: start vs end of drought

Same dry WY repeated every drought year. Start = first dry year after average spinup (index 40). End = last dry year (index 40 + L - 1). Members: 3_year_drought, 10_year_drought.

Domain	Drought length	Start NSE / KGE	End NSE / KGE	Start->end PBIAS
Potomac	3 yr	0.19 / 0.52	0.23 / 0.51	-29% -> -32%
Potomac	10 yr	0.19 / 0.52	0.23 / 0.50	-29% -> -33%
Wolf	3 yr	-0.28 / 0.17	-0.11 / 0.42	-28% -> -52%
Wolf	10 yr	-0.28 / 0.17	-0.11 / 0.41	-28% -> -53%

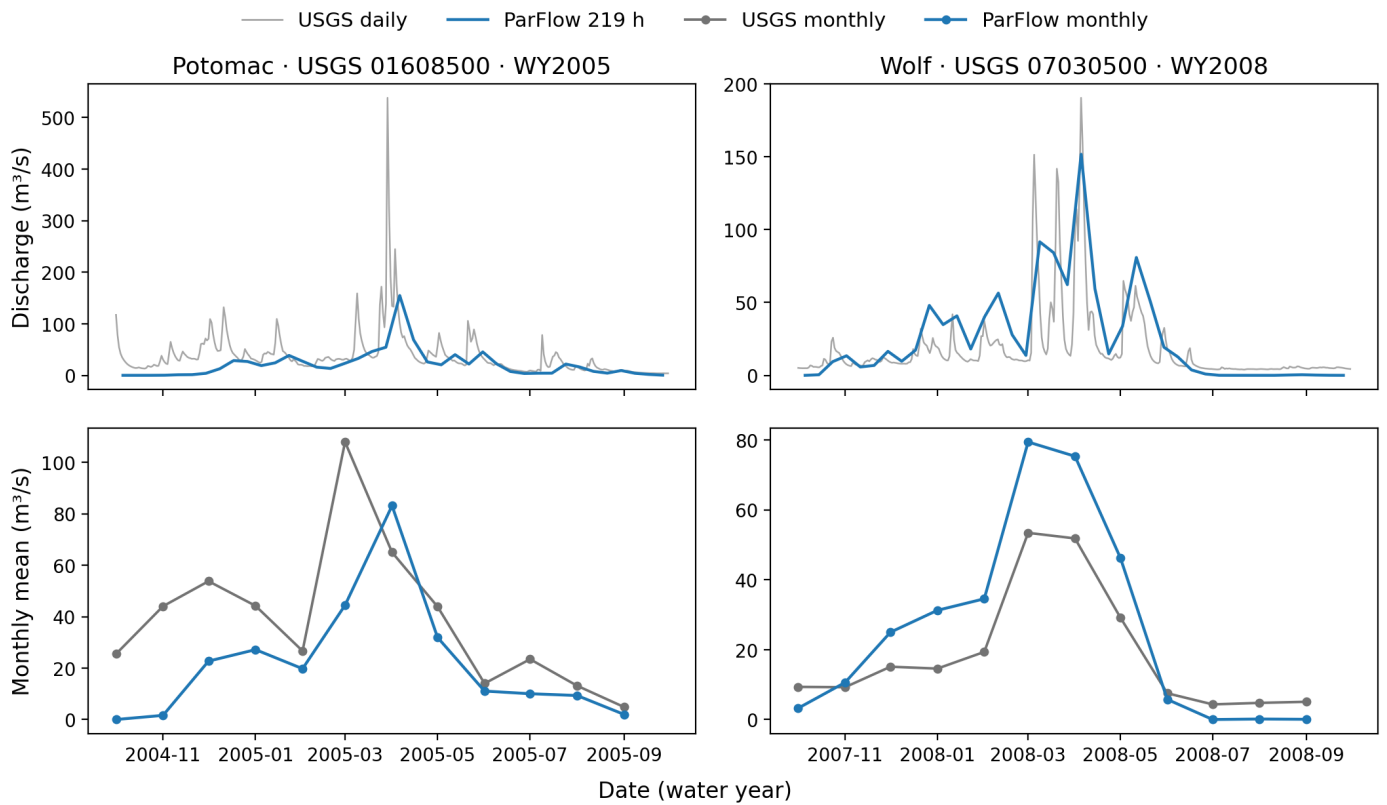
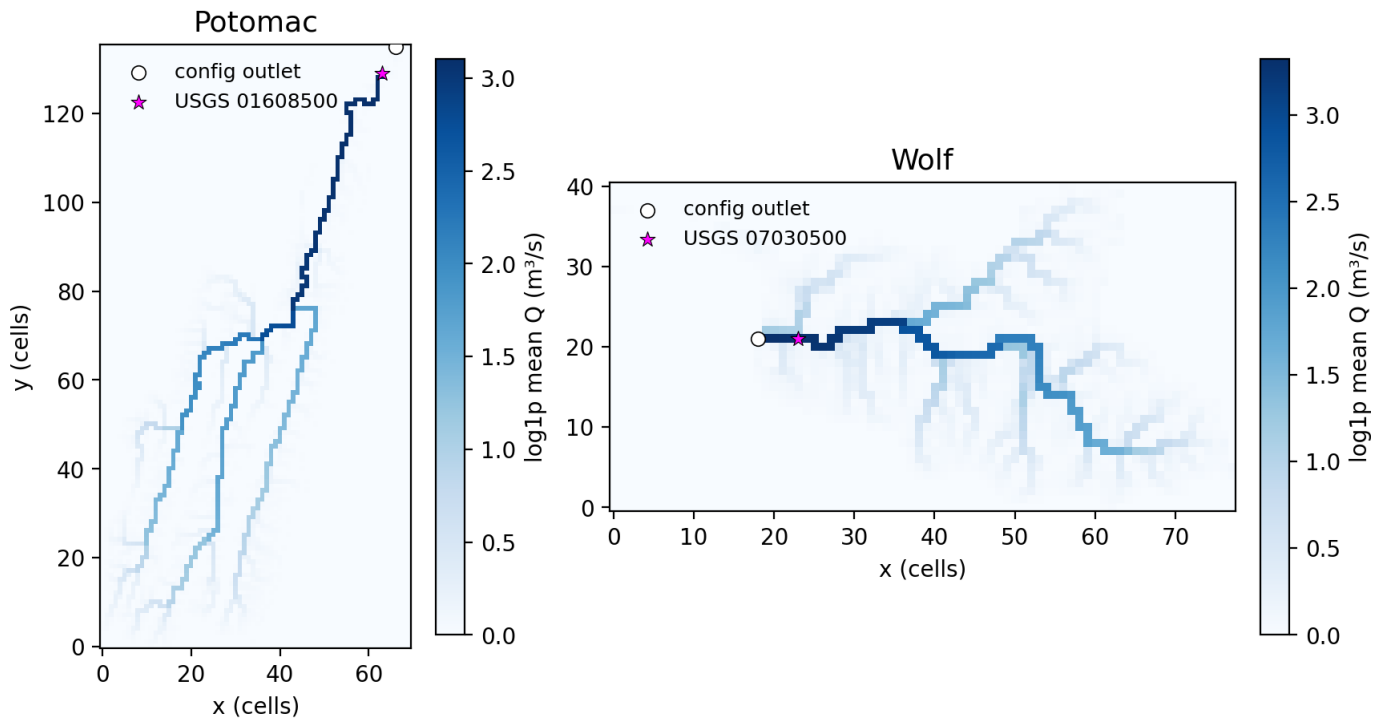
Potomac barely changes after prior dry years. Wolf dries further (more negative PBIAS) while KGE rises; 3-year end ~ 10-year end, so most of the memory effect appears within the first few dry years.

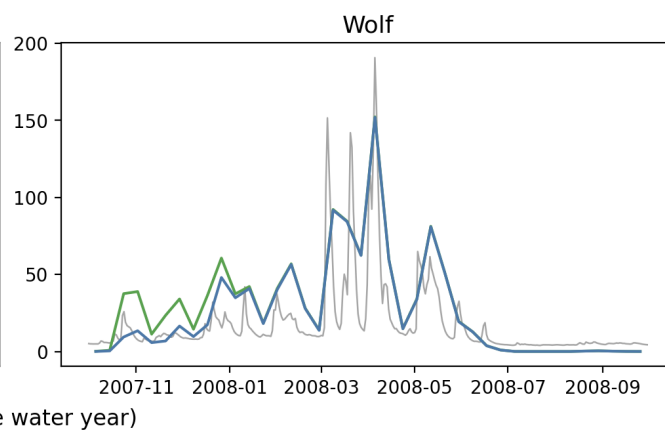
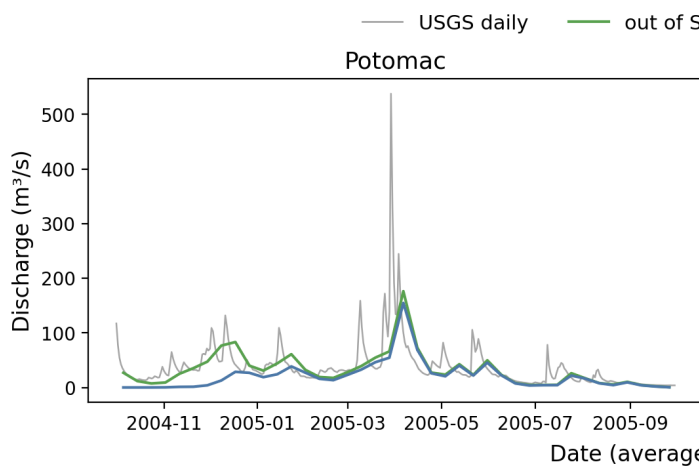
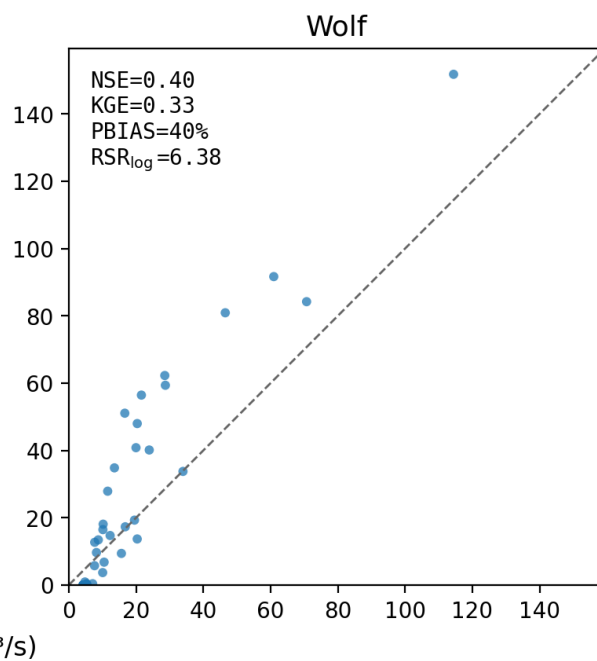
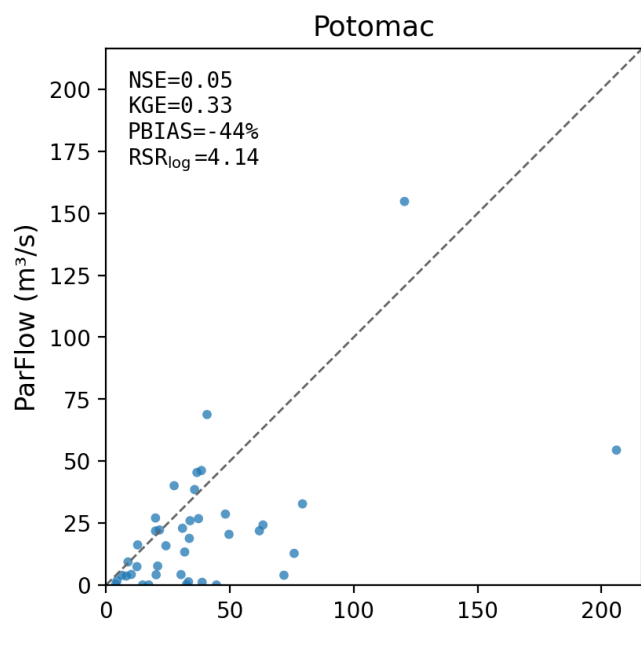
Takeaways

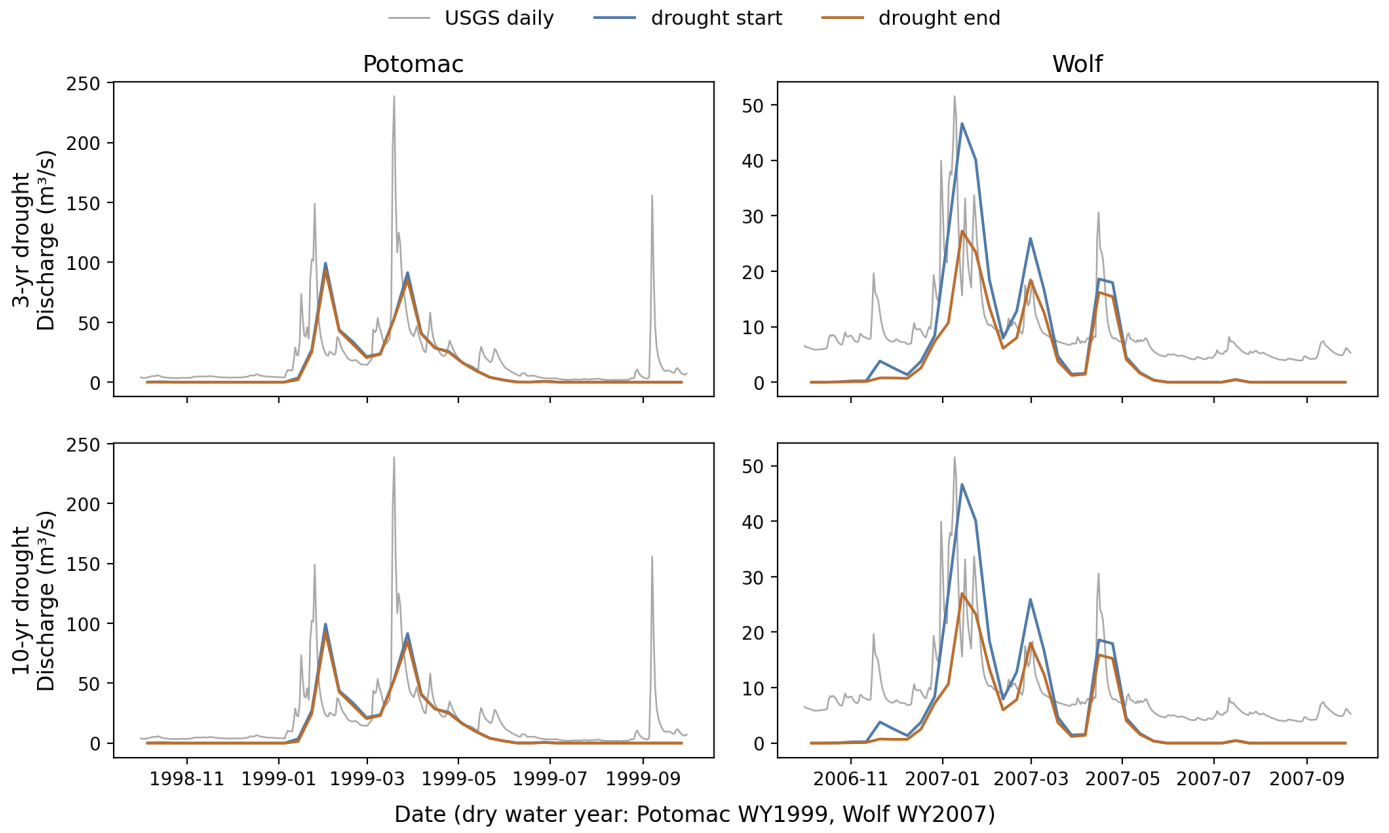
1. Gage choice matters - use Springfield (01608500) for Potomac outlet-scale checks, not Brandywine.
2. Antecedent conditions matter as much as the forcing year - average-year skill differs strongly y0 vs y39; dry-year skill (especially Wolf) differs drought start vs end.
3. Domains behave differently - Potomac validates better out of SS and degrades over average spinup; Wolf validates better after spinup and is sensitive to multi-year drought memory.
4. Spun-up average-year Q is biased low (Potomac) / high (Wolf) by tens of percent vs USGS; treat ensemble streamflow anomalies relative to the model baseline rather than as absolute USGS matches unless bias-corrected.

How to reproduce

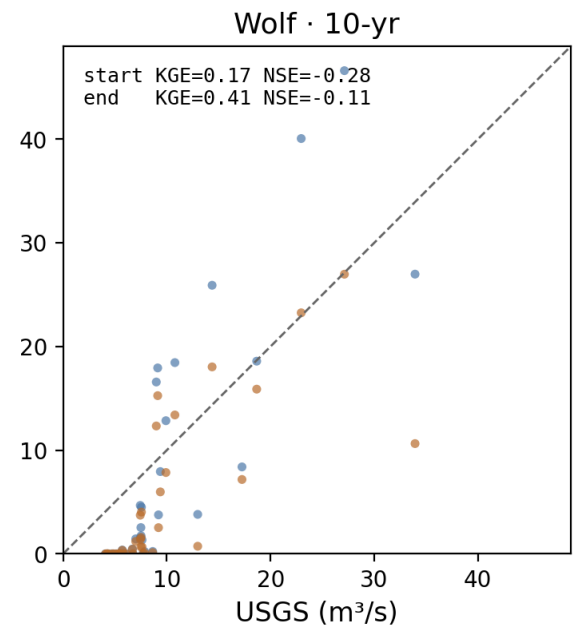
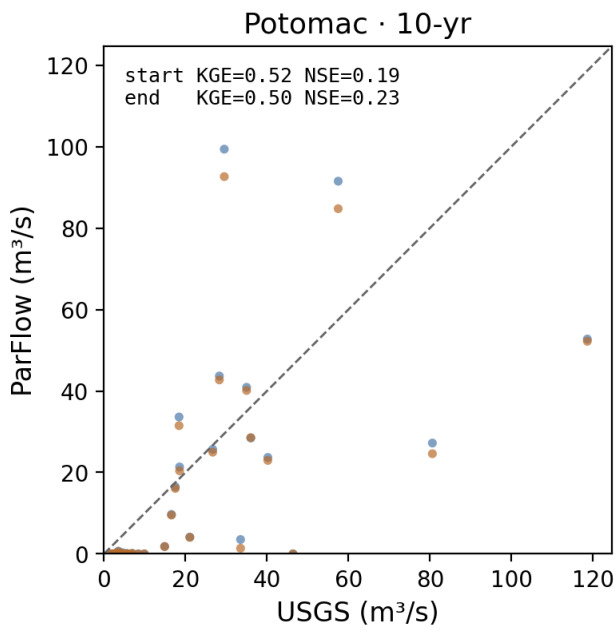
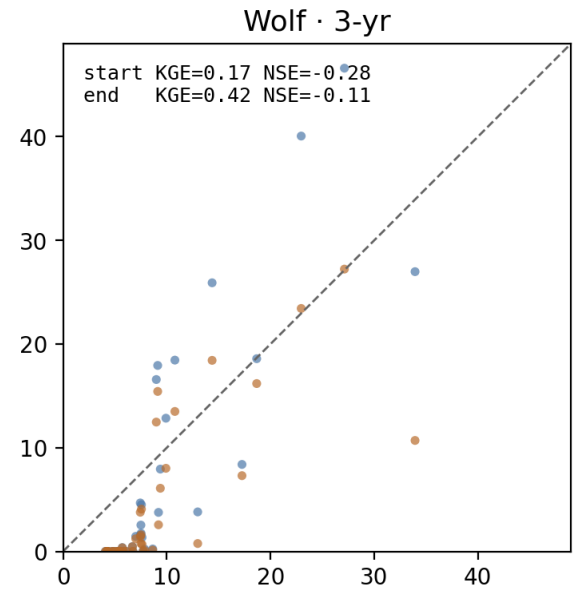
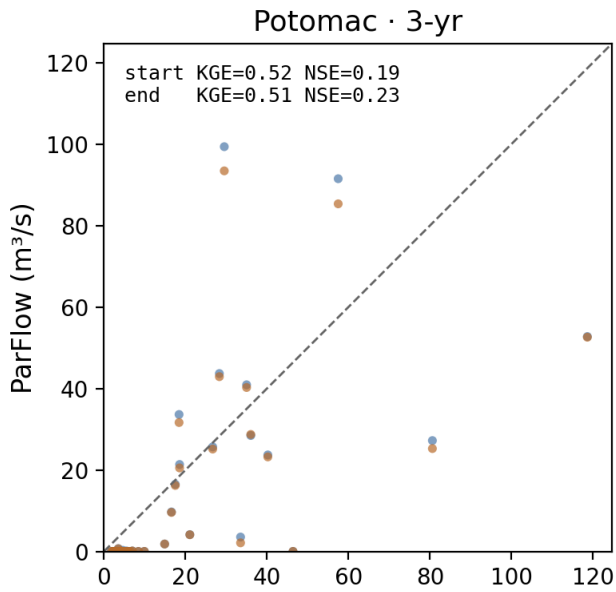
```
```bash
conda activate /glade/work/bwest/conda-envs/droughts
cd /glade/derecho/scratch/bwest/drought-ensemble
jupyter notebook analysis/average_year_usgs_validation.ipynb
```







● start ● end



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